

Notes: Petersen and Rajan (RFS, 1997)

Trade Credit: Theories and Evidence

Contents

1	Big picture	3
2	Incremental contribution of the paper	3
3	Data and measurement	4
3.1	Theoretical channels	4
3.2	Main data source: NSSBF small firms	4
3.3	Large-firm comparison sample	5
3.4	Key trade-credit variables	5
4	Methodology and empirical specifications	6
4.1	Balance-sheet mapping	6
4.2	Who offers credit? Accounts receivable regressions	6
4.3	Who is offered credit? Purchases made on account	6
4.4	Who uses credit? Accounts payable regressions	7
4.5	How the paper separates supply and demand	8
5	Identification and interpretation	8
5.1	What the paper can identify	8
5.2	Why accounts receivable are useful	8
5.3	Why accounts payable are harder to interpret	8
5.4	Supplier financing advantage versus bank substitution	9
5.5	Price discrimination and relationship value	9
5.6	Liquidation and control	9
5.7	Limits of the evidence	9
6	Tables and figures	9
6.1	Figure 1 and Table 1: Balance-sheet mapping and trade-credit use by firm size	9
6.2	Table 2: Trade credit offered and used by small firms	10
6.3	Table 3: Determinants of accounts receivable	11
6.4	Table 4: Summary statistics for the receivables regression	11

6.5	Table 5: Trade-credit supply	12
6.6	Table 6: Trade-credit demand	13
7	Short summary	14
8	Conclusion	15
8.1	Core conclusion	15
8.2	Economic intuition	15
8.3	Incremental contribution revisited	15
8.4	Implications	15
8.5	Limitations	16
8.6	Takeaway	16

1 Big picture

- **Research question.** Why do firms borrow from and lend to other firms through trade credit when specialized financial institutions also provide credit? More specifically, who offers trade credit, who receives it, and what theories explain these patterns?
- **Main idea.** Trade credit is not just a passive accounting item. It is a form of short-term financing that travels through supplier-customer relationships. The paper studies whether suppliers have financing, information, control, liquidation, price-discrimination, or transaction-cost advantages that explain why they extend credit.
- **Empirical object.** The paper studies two sides of the firm's balance sheet: accounts receivable, which proxy for trade credit the firm extends to customers, and accounts payable, which proxy for trade credit the firm receives from suppliers.
- **Main finding.** Firms with better access to capital markets extend more trade credit, while firms with weaker access to institutional finance rely more on trade credit. This supports the view that trade credit redistributes financing capacity from firms with better capital-market access to firms with worse access.
- **Supplier advantage.** The evidence suggests that suppliers can sometimes lend when banks will not because suppliers observe product-market information, can threaten to stop future supplies, and may have an advantage in repossessing or reselling goods.
- **Strategic use.** Trade credit also appears related to price discrimination and customer relationships. Firms with high gross margins and suppliers selling goods that are easier to liquidate are more likely to extend credit.
- **Demand side.** A firm's use of trade credit is strongly related to the predicted supply of trade credit from its suppliers, its demand for funds, and its access to alternative credit.
- **Important distinction.** Accounts payable are not a pure demand variable. They reflect both how much credit suppliers are willing to offer and how much the firm chooses to use. The paper therefore tries to separate supply from demand by first estimating credit purchases offered by suppliers and then using the predicted supply in accounts-payable regressions.

Economic takeaway. Trade credit exists because suppliers are sometimes better positioned than banks to finance particular customers. They can use information from repeated product-market transactions, can discipline buyers by withholding future deliveries, and can recover value from goods more effectively. At the same time, constrained firms demand trade credit because it substitutes for scarce bank or market financing.

2 Incremental contribution of the paper

- **Unifies several theories.** Earlier theories explain trade credit through supplier financing advantages, price discrimination, and transaction costs. Petersen and Rajan organize these theories and ask which ones line up with detailed firm-level evidence.
- **Uses small-firm microdata.** The paper exploits the National Survey of Small Business Finances, which contains detailed information on small firms' accounts receivable, accounts payable, credit purchases, bank relationships, credit denials, lines of credit, and firm characteristics.

- **Compares small and large firms.** By combining the small-firm survey with Compustat data for large firms, the paper shows that large firms both borrow and lend more trade credit than small firms, but small firms rely on trade credit in ways that reveal financing frictions.
- **Separates offered credit from used credit.** A central empirical contribution is the attempt to distinguish trade-credit supply from trade-credit demand. Purchases made on account proxy for what suppliers offer, while accounts payable reflect both the offered amount and repayment speed.
- **Tests specific supplier advantages.** The paper does not merely ask whether trade credit exists. It tests whether credit supply is stronger when suppliers likely have an advantage in monitoring, control, liquidation, and relationship-specific customer valuation.
- **Links product markets and finance.** The paper shows that the availability of trade credit depends not only on financial variables but also on product-market variables such as gross margins, sales growth, industry, inventory composition, and customer-supplier relationships.
- **Clarifies the role of banks.** The evidence suggests that trade credit and bank credit can be substitutes for constrained firms, but the relationship is not mechanically negative. Firms with more bank access may also be larger or better quality, and therefore more involved in credit relationships overall.
- **Provides a foundation for later network and contagion work.** By treating trade credit as a set of financial links between firms, the paper anticipates later work on trade-credit chains, supplier-customer networks, and the propagation of shocks.

3 Data and measurement

3.1 Theoretical channels

The paper begins by grouping theories of trade credit into three broad families.

- **Financing advantage theories.** Suppliers may have an advantage over financial institutions in lending to their customers because they observe the buyer’s business, control a key input, or can reclaim and resell goods more easily.
- **Price-discrimination theories.** Trade credit can be a way for a supplier with market power to offer an implicit discount to selected customers without changing the posted product price.
- **Transaction-cost theories.** Trade credit can reduce the need to pay every invoice immediately and can help firms manage the timing of cash payments and receipts.

Interpretation. These theories are not mutually exclusive. A supplier may extend credit because it can monitor the customer, because it wants to preserve a valuable relationship, and because trade credit is administratively convenient.

3.2 Main data source: NSSBF small firms

The principal small-firm data source is the National Survey of Small Business Finances. The survey covers nonfinancial, nonfarm, for-profit small businesses with fewer than 500 employees.

- **Sample.** The survey contains detailed information for roughly 3,400 firms, although the usable sample varies by specification.
- **Firm scale.** The firms are small: the median firm has book assets of about \$130,000 and only a small number of employees.
- **Balance-sheet variables.** The survey reports accounts receivable, accounts payable, assets, sales, profits, and other accounting variables.
- **Trade-credit terms.** The survey reports the fraction of purchases made on account and information about early-payment discounts.
- **Bank relationships.** The survey reports credit lines, loan denials, loan applications, the length of the firm's relationship with its lender, and the risk premium on the most recent loan.

3.3 Large-firm comparison sample

To compare small and large firms, the paper uses Compustat data for public firms. This large-firm sample helps document how accounts payable and accounts receivable differ by size and industry.

Interpretation. The comparison shows that larger firms both offer and use more trade credit. However, this does not mean that large firms are more constrained. It also reflects scale, deeper product-market networks, and greater capacity to lend to customers.

3.4 Key trade-credit variables

- **Accounts receivable over sales.** This measures trade credit extended by the firm to its customers. It is the main proxy for who offers credit.
- **Accounts payable over assets.** This measures trade credit received from suppliers and still unpaid. It is the main proxy for who uses credit, but it reflects both credit supply and repayment timing.
- **Purchases made on account over assets.** This proxies for how much trade credit suppliers offer to the firm. It is built from the fraction of purchases made on account and the firm's cost of goods sold.
- **Available line of credit.** This proxies for access to bank financing.
- **Loan denial and loan application variables.** These capture unmet or attempted demand for institutional credit.
- **Gross profit margin.** This proxies for potential price-discrimination incentives and the value of preserving customer relationships.
- **Finished-goods inventory share.** This proxies for liquidation costs: finished goods are less useful to suppliers than raw materials or intermediate goods.

4 Methodology and empirical specifications

4.1 Balance-sheet mapping

Figure 1 is the conceptual map. The firm's suppliers provide inputs and may allow delayed payment. This appears as accounts payable on the firm's balance sheet. The firm may also sell to customers on credit. This appears as accounts receivable on the firm's balance sheet.

$$\text{Supplier credit to firm} \rightarrow AP_i, \quad \text{Firm credit to customers} \rightarrow AR_i. \quad (1)$$

Interpretation. The same firm can be both a borrower and a lender. This is why trade credit creates credit chains: one firm's payable is another firm's receivable.

4.2 Who offers credit? Accounts receivable regressions

The first major empirical exercise studies which firms extend more credit to customers. A simplified version of the accounts-receivable specification is:

$$\frac{AR_i}{Sales_i} = \alpha + X_i' \beta + \varepsilon_i, \quad (2)$$

where X_i includes firm size, age, available credit line, profits, sales growth, gross profit margin, metropolitan location, industry indicators, and related controls.

- Larger firms extend more trade credit.
- Older firms tend to extend more credit, though the relation is nonlinear.
- Firms with larger available lines of credit extend more credit, consistent with a financing-capacity channel.
- Gross profit margins are positively related to accounts receivable at lower levels but the squared term is negative, implying a concave relation.
- Firms located in metropolitan statistical areas tend to extend more credit, consistent with better enforcement or more active commercial networks.

Interpretation. Suppliers that can raise funds more easily are better able to finance customers. This is the core evidence for the idea that trade credit redistributes financing from firms with better credit access to firms with worse credit access.

4.3 Who is offered credit? Purchases made on account

The second major empirical exercise studies what kinds of firms suppliers are willing to finance. The dependent variable is credit purchases over assets:

$$\frac{CreditPurchases_i}{Assets_i} = \alpha + Z_i' \theta + u_i. \quad (3)$$

This measure is meant to capture trade-credit supply from suppliers, before repayment timing turns it into accounts payable.

- Firms with more employees relative to assets receive more trade credit, which partly reflects purchases and production needs.
- More profitable firms receive more trade credit, consistent with suppliers lending to better-quality customers.
- Incorporated firms receive more trade credit.
- Supplier-related variables matter: trade credit is higher for firms where suppliers have stronger reasons to preserve or finance relationships.
- The share of inventory that is finished goods enters negatively. Suppliers appear more willing to extend credit when the goods they supply are easier to recover or resell.

Interpretation. Trade-credit supply is not random. Suppliers extend credit when the buyer looks creditworthy, when the relationship is valuable, and when the supplier can better control or liquidate the goods.

4.4 Who uses credit? Accounts payable regressions

The third major empirical exercise studies accounts payable over assets. A simplified version is:

$$\frac{AP_i}{Assets_i} = \alpha + \gamma \widehat{Supply}_i + D_i' \delta + e_i, \quad (4)$$

where $\widehat{Supply}_i$ is predicted trade-credit supply from the purchases-on-account model, and D_i includes variables measuring demand for capital and credit availability.

- Predicted supply of trade credit is positive and significant. Firms use more trade credit when suppliers offer more of it.
- Positive sales growth is associated with more trade-credit use, consistent with demand for funds during expansion.
- Lower profitability is associated with more accounts payable, consistent with trade credit being used when internal funds are weak.
- More available bank credit is associated with less trade-credit use, suggesting substitution between bank credit and trade credit for some firms.
- Longer relationships with lenders are associated with lower accounts payable, again consistent with bank relationships reducing reliance on supplier credit.
- Early-payment discount variables help probe the price of trade credit, though the price evidence is weaker than the quantity evidence.

Interpretation. Accounts payable reflect both opportunity and need. Suppliers must offer trade credit, but firms rely on it more when they are growing, have weaker internal funds, or lack alternative institutional credit.

4.5 How the paper separates supply and demand

A key challenge is that accounts payable combine supply and demand. Suppose a firm has high accounts payable. This could mean suppliers generously offer credit, or it could mean the firm delays repayment because it is financially strained.

The paper's two-step logic is:

- First, estimate trade-credit supply using purchases made on account.
- Second, use predicted supply as an explanatory variable in the accounts-payable regression.
- Then interpret the remaining variables as demand or repayment-timing factors, conditional on available trade credit.

Interpretation. This is not a perfect structural separation, but it is a useful way to avoid treating accounts payable as a pure demand measure.

5 Identification and interpretation

5.1 What the paper can identify

The paper identifies robust cross-sectional relationships between trade credit and firm characteristics. It can show that firms with better access to finance extend more credit and that firms with weaker access use more trade credit. It is less able to establish clean causal effects, because firm quality, product-market relationships, and financing access are jointly determined.

5.2 Why accounts receivable are useful

Accounts receivable are a natural measure of credit extended to customers. They capture the amount owed to the firm by its buyers and therefore reveal the supplier side of the credit relationship.

The limitation is that receivables also depend on customer demand, payment speed, industry terms, and collection technology. A firm may have low receivables because it does not offer credit, because customers pay quickly, or because the industry uses cash transactions.

5.3 Why accounts payable are harder to interpret

Accounts payable are even more ambiguous. A firm with high payables may receive generous trade-credit supply, may have high purchases, may be delaying payment because of distress, or may simply operate in an industry where payment terms are long.

This is why the paper studies purchases made on account and credit terms in addition to payables.

5.4 Supplier financing advantage versus bank substitution

The paper's interpretation is that suppliers can sometimes lend more effectively than banks. But this does not mean suppliers are always cheaper lenders. Trade credit can be expensive, especially when early-payment discounts are foregone. The key point is availability: constrained firms may use supplier credit because it is available when bank credit is not.

5.5 Price discrimination and relationship value

High gross profit margins and strong customer relationships can make it worthwhile for suppliers to finance buyers. A supplier may prefer to preserve a profitable customer rather than cut off credit immediately. This mechanism is closer to price discrimination or relationship investment than to pure financing advantage.

5.6 Liquidation and control

Suppliers may be better than banks at recovering value because they know the goods, can repossess goods, can stop future deliveries, or can redirect goods to other customers. The negative relation between finished-goods inventory and credit supply supports this idea: suppliers appear more willing to lend when the goods are easier to control or liquidate.

5.7 Limits of the evidence

The evidence is cross-sectional and relies on proxies. Some variables, such as available credit lines, profitability, and bank relationships, are endogenous. The NSSBF is rich but focuses on small firms at one point in time, while the Compustat comparison covers large public firms with different reporting and business models.

6 Tables and figures

6.1 Figure 1 and Table 1: Balance-sheet mapping and trade-credit use by firm size

Read Figure 1:

- The firm's suppliers extend credit to the firm. This becomes accounts payable for the firm.
- The firm extends credit to its customers. This becomes accounts receivable for the firm.
- The diagram clarifies that trade credit has two directions: the firm is a borrower on the input side and a lender on the output side.

Read Table 1:

- The table compares accounts payable and accounts receivable for small NSSBF firms and large Compustat firms by industry.

- Large firms have much higher accounts-payable-to-sales and accounts-receivable-to-sales ratios than small firms.
- For small firms, the total mean accounts payable over sales is 4.4%, while accounts receivable over sales is 7.3%.
- For large firms, the corresponding totals are 11.6% and 18.5%.
- The fraction of firms with zero trade credit is much higher among small firms.

Economic message: Trade credit is widespread but not universal. Larger firms are more deeply embedded in credit relationships, both as borrowers from suppliers and as lenders to customers.

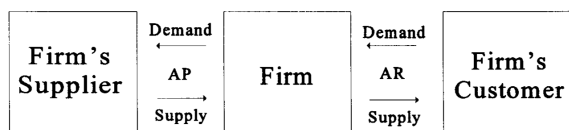


Figure 1
The trade credit relationships
The amount of trade credit extended between the firm and its suppliers will appear as the accounts payable on the balance sheet of the firm. The amount of trade credit extended between the firm and its customers will appear as accounts receivable on the balance sheet of the firm.

Figure 1: Trade credit relationships.

Table 1
Accounts payables and receivables to sales ratios

Industry	Accounts payable/sales			Accounts receivable/sales		
	Mean	Median	% Zero	Mean	Median	% Zero
Panel A: Small firms^a						
Mining	6.1	4.6	23.1	9.9	6.9	30.8
Construction	5.4	2.5	25.7	10.4	7.8	14.8
Manufacturing	6.5	4.2	11.7	11.8	10.0	9.8
Transportation/utilities	3.8	1.9	31.5	8.1	6.5	17.6
Wholesale trade	7.0	3.8	15.6	8.1	7.0	6.2
Retail trade	3.9	1.7	26.6	3.0	0.4	39.6
Services	2.7	0.0	51.9	8.0	3.5	34.2
Total	4.4	1.8	30.9	7.3	5.8	26.5
Panel B: Large firms^b						
Mining	25.2	17.5	0.5	28.7	21.7	2.0
Construction	17.0	8.3	0.0	15.8	16.4	23.7
Manufacturing	9.8	7.4	0.4	19.1	17.0	0.6
Transportation/utilities	14.3	8.6	1.2	16.2	14.1	2.6
Wholesale trade	12.5	8.6	2.0	15.5	14.0	0.9
Retail trade	8.6	6.7	0.8	7.3	2.3	6.2
Services	10.6	6.5	1.1	22.4	19.4	3.7
Total	11.6	7.6	0.7	18.5	16.1	2.3

^aThis table is based on the National Survey of Small Business Finance. Firms in the agricultural (SIC 100–999) and financial sectors (SIC 6000–6999) were excluded. Ratios are expressed as percents.
^bThis table is based on the Compustat database. Firms in the agricultural (SIC 100–999) and financial sectors (SIC 6000–6999) were excluded. Ratios are expressed as percents.

Table 1: Accounts payable and accounts receivable to sales ratios.

6.2 Table 2: Trade credit offered and used by small firms

Read:

- Table 2 groups NSSBF firms by book-asset percentile and industry.
- Panel A reports accounts receivable over sales.
- Panel B reports accounts payable over assets.
- Panel C reports credit purchases over assets, the paper’s proxy for the supply of trade credit from suppliers.
- Across many industries, receivables and payables rise with firm size.
- Credit purchases over assets vary substantially across industries, especially between wholesale, retail, construction, manufacturing, transportation, and services.

Economic message: Even within small firms, firm size and industry matter a lot. Trade credit is not a uniform contract; it reflects production technology, customer-supplier relationships, and financing access.

6.3 Table 3: Determinants of accounts receivable

Read:

- The dependent variable is accounts receivable over sales.
- Larger firms extend more trade credit.
- Older firms tend to extend more credit, but the age effect is concave in some specifications.
- A larger available line of credit predicts more accounts receivable, indicating that firms with better access to finance can lend more to customers.
- Gross profit margins are positively related to receivables, while the squared term is negative, consistent with a concave price-discrimination or relationship-value effect.
- Firms located in metropolitan statistical areas extend more credit in several specifications.

Economic message: Firms that can finance credit and that benefit more from preserving customer relationships offer more trade credit. This supports both financing-advantage and strategic-pricing views.

Table 2
Trade credit offered and used by firms (by size and by industry)

Book value of assets (\$1000)	Asset percentile	Industry						
		Construction and mining	Manufacturing	Transportation	Wholesale (durables)	Wholesale (nondurables)	Retail	Services
Panel A: Accounts receivable ^a								
Less than 15	0-10	0.00 (42)	0.00 (17)	0.00 (5)	0.00 (7)	0.02 (5)	0.00 (54)	0.00 (133)
15-40	10-25	0.05 (86)	0.02 (30)	0.00 (7)	0.05 (18)	0.04 (12)	0.00 (147)	0.02 (167)
40-130	25-50	0.07 (86)	0.06 (52)	0.03 (28)	0.06 (36)	0.06 (24)	0.00 (288)	0.04 (244)
130-488	50-75	0.09 (102)	0.10 (95)	0.07 (53)	0.08 (47)	0.06 (56)	0.01 (230)	0.05 (215)
488-2293	75-90	0.12 (54)	0.12 (86)	0.08 (22)	0.09 (43)	0.05 (40)	0.01 (120)	0.09 (83)
Over 2293	90-100	0.17 (33)	0.12 (106)	0.10 (13)	0.11 (25)	0.06 (28)	0.01 (37)	0.14 (42)
Panel B: Accounts payable to assets ^b								
Less than 15	0-10	0.00 (42)	0.00 (17)	0.00 (5)	0.02 (12)	0.00 (54)	0.00 (133)	
15-40	10-25	0.02 (86)	0.02 (30)	0.00 (7)	0.12 (30)	0.02 (147)	0.00 (167)	
40-130	25-50	0.05 (86)	0.05 (52)	0.00 (28)	0.07 (60)	0.03 (288)	0.00 (244)	
130-488	50-75	0.08 (102)	0.11 (95)	0.02 (53)	0.12 (83)	0.07 (230)	0.00 (215)	
488-2293	75-90	0.14 (54)	0.14 (86)	0.04 (22)	0.16 (83)	0.09 (120)	0.03 (83)	
Over 2293	90-100	0.18 (33)	0.12 (106)	0.05 (13)	0.17 (53)	0.06 (57)	0.05 (42)	
Panel C: Credit purchases to assets ^c								
Less than 15	0-10	2.54 (45)	2.01 (12)	0.00 (2)	2.99 (9)	1.31 (28)	0.00 (70)	
15-40	10-25	1.38 (86)	1.10 (18)	0.00 (4)	1.77 (21)	1.02 (98)	0.00 (117)	
40-130	25-50	1.36 (82)	0.91 (47)	0.00 (18)	1.26 (49)	1.24 (200)	0.00 (169)	
130-488	50-75	0.98 (93)	1.25 (82)	0.00 (29)	2.07 (72)	1.36 (193)	0.00 (158)	
488-2293	75-90	1.55 (47)	1.42 (70)	0.00 (25)	2.00 (75)	1.43 (109)	0.00 (68)	
Over 2293	90-100	1.85 (27)	1.35 (89)	0.00 (11)	2.03 (48)	1.85 (49)	0.00 (41)	

^aThis panel describes the median accounts receivable-to-sales ratio for firms in the NSSB. The number of firms in each cell is reported in parentheses.

^bThis panel describes the median accounts payable to assets ratio for firms in the NSSB. The number of firms in each cell is reported in parentheses.

^cCredit purchases are defined as the product of cost of goods sold and the fraction purchased on account. The panel describes the median accounts payable to assets ratio for firms in the NSSB. The number of firms in each cell is reported in parentheses.

^aThis panel describes the median accounts receivable:assets ratio for firms in the NSSRF. The number of firms in each cell is reported in parentheses.
^bThis panel describes the median accounts payable to assets ratio for firms in the NSSRF. The number of firms in each cell is reported in parentheses.
^cCredit purchases are defined as the product of cost of goods sold and the fraction purchased on account. The panel describes the median purchases:assets ratio for firms in the NSSRF. The number of firms in each cell is reported in parentheses.

Table 2: Trade credit offered and used by small firms.

Table 3
The determinants of accounts receivable

Independent variable	I	II	III	IV ^a	V ^b
Log(book value of assets)	.013* (.001)	.013* (.001)	.014* (.001)	.012* (.001)	.010* (.003)
Log(1 + firm age) (in years)	.006* (.002)	.024** (.010)	.025* (.010)	.013 (.009)	.033 (.021)
Log(1 + firm age) ² (in years)		-.004** (.002)	-.004** (.002)	-.005 (.002)	-.007 (.004)
Maximum available line of credit/sales	.027* (.006)	.027* (.006)	.026* (.006)	.021* (.005)	.103** (.051)
Net profits/sales	-.031* (.010)	-.033* (.010)			
Net profits/sales if positive, zero otherwise			-.004 (.013)	-.009 (.012)	.023 (.042)
Net profits/sales if negative, zero otherwise			-.101* (.021)	-.105* (.020)	-.153* (.054)
Percent sales growth (86-87) if positive, zero otherwise	.032* (.010)	.033* (.010)	.033* (.010)	.004 (.009)	-.002 (.027)
Percent sales growth (86-87) if negative, zero otherwise	-.051* (.009)	-.053* (.009)	-.048* (.009)	-.039* (.008)	-.039 (.028)
Gross profit margin/sales	.016** (.008)	.060** (.028)	.056** (.028)	.117* (.030)	.201** (.079)
(Gross profit margin/sales) ²		-.043*** (.027)	-.045*** (.026)	-.118* (.030)	-.206** (.096)
Firm is in MSA (0,1)	.014* (.004)	.014* (.004)	.014* (.004)	.007** (.003)	.012 (.009)
Wholesaler of durable goods (0,1)					.001 (.009)
Number of observations	1805	1805	1805	1805	277
Adjusted R ²	0.141	0.144	0.150	0.315	0.154

The dependent variable is the accounts receivable:sales ratio reported by the firm. The coefficients are estimated using ordinary least squares. Standard errors are in parentheses. Each regression has a constant whose coefficient is not reported. Firms in the financial industry (SIC 6000-6999) and the service industry (SIC 7000-8999) are excluded unless otherwise stated. When the distribution of a variable was highly skewed for high values, we recoded the highest percent of values to the 99th percentile of the distribution. Similarly if it was highly skewed for low values, we recoded the lowest percent of values to the 1st percentile of the distribution. These variables include accounts receivable to sales, net profits to sales, sales growth, and gross profit margin to sales.

^a40 two-digit SIC indicators are included in addition to the constant.

^bIncludes only wholesalers (SIC codes 5000-5199).

*, **, ***Coefficient significantly different from zero at the 1%, 5%, and 10% level, respectively.

Table 3: Determinants of accounts receivable.

6.4 Table 4: Summary statistics for the receivables regression

Read:

- Table 4 summarizes variables used in the accounts-receivable regressions.
- The mean accounts-receivable-to-sales ratio is 0.066, and the median is 0.039.

- The average available line of credit over sales is 0.06, while the median is zero.
- Gross profits over sales average 0.40, with a median of 0.35.

Economic message: The table shows substantial heterogeneity in trade credit and financing capacity. Many small firms have no available line of credit, making supplier financing potentially important.

6.5 Table 5: Trade-credit supply

Read:

- The dependent variable is purchases made on credit over assets.
- Employees over assets enters positively, reflecting production and purchase needs.
- Profitability enters positively: more profitable firms are offered more trade credit.
- Incorporation status is positively associated with trade-credit supply.
- Long relationships with financial institutions and loan-denial variables do not explain trade-credit supply strongly.
- The percent of inventory that is finished goods enters negatively. Suppliers extend less credit when inventory is harder for them to recover or liquidate.
- The interaction-style profit variables suggest that suppliers distinguish between unprofitable growing firms and unprofitable declining firms.

Economic message: Suppliers do not simply lend to the weakest firms. They extend credit when the buyer looks viable and when the supplier has control or liquidation advantages. This is strong evidence for the supplier-advantage theory.

Table 4
Summary statistics

Variables	Mean	Median	Standard deviation
Accounts receivable:sales ratio	0.066	0.039	0.083
Log(book value of assets)	12.18	11.96	1.93
Log (1 + firm age) (in years)	2.35	2.39	0.88
Maximum available line of credit/sales	0.06	0.00	0.09
Net profit/sales	0.10	0.05	0.23
Percent sales growth (86–87) if positive, zero otherwise	0.26	0.07	0.59
Percent sales growth (86–87) if negative, zero otherwise	−0.04	0.00	0.11
Gross profits/sales	0.40	0.35	0.25

Table 4: Summary statistics.

Table 5
Trade credit supply: Purchases made on credit over assets

Independent variable	I	II ^b	III	IV	V
Proxy for wages					
Employees/assets (\$10,000)	.200* (.042)	.247* (.041)	.366* (.068)	.199* (.042)	.174* (.043)
Firm credit quality					
Log(book value of assets)	.049 (.033)	.042 (.032)	.105** (.048)	.051 (.033)	.060*** (.033)
Net profits/assets	.326* (.058)	.326* (.055)	.420* (.092)	.325* (.058)	
Log(1+firm age)	.044 (.196)	.073 (.186)	−.353 (.272)	.044 (.196)	.078 (.195)
Log(1+firm age) ²	−.029 (.043)	−.039 (.041)	.031 (.061)	−.030 (.043)	−.037 (.043)
Firm is incorporated	.476* (.108)	.510* (.103)	.221 (.153)	.481* (.108)	.452* (.108)
Relationships with financial institutions					
Longest relationship with lender (in years)	−.015 (.066)	−.017 (.063)	.121 (.098)	−.018 (.066)	.003 (.065)
Risk premium on most recent loan (premium over treasuries)			−.015 (.024)		
Firm denied request for loan in the last year (0,1)				−.213 (.203)	
Relationships with suppliers					
Net profits/assets (if profits > 0)					.422* (.064)
Net profits/assets (if profits < 0 and sales growth > 0)					−.844* (.275)
Net profits/assets (if profits < 0 and sales growth < 0)					.228 (.433)
Liquidation costs ^a					
Percent of inventory which is finished goods	−.725* (.167)		−.661* (.229)	−.717* (.167)	−.709* (.166)
Number of observations	1644	1706	758	1644	1644
R ²	12.0	21.2	15.4	12.0	13.0

The dependent variable is the firm's purchases on account divided by assets. Purchases on account is calculated as the percent of purchases made on account times the firm's costs of goods sold. The coefficients are estimated using ordinary least squares. Standard errors are in parentheses. Each regression includes one-digit SIC industry dummy variables (unless otherwise specified) and a constant whose coefficients are not reported. Firms in the financial industry (SIC 6000–6999) and service industry (SIC 7000–8999) are excluded unless otherwise stated. Due to low asset values 23 firms had extremely high ratios for purchase on account to assets. These observations were recoded to the 99th percentile of the distribution. We tested this approach by estimating the regression using a tobit with an upper limit of the 99th percentile of the purchase on account regression. The empirical results are qualitatively similar. Low asset values also caused some of the independent variables to be skewed. When the distribution of a variable was highly skewed, we recoded the upper percent of values to the 99th percentile of the distribution. These variables include operating profits to assets, employees to assets, and number of trade creditors to assets. ^aIndustry variables are calculated using 1987 Compustat data. Each variable was calculated for firms in a two-digit SIC code industry. These average industry values were then matched to the firms in the NSSBF. ^b This regression includes 40 industry dummies, one for each two-digit SIC code in our dataset. The percent of inventory which is finished goods is based on industry averages and must therefore be dropped from this specification.

*, **, ***Coefficient significantly different from zero at the 1%, 5%, and 10% levels, respectively.

Table 5: Trade-credit supply: purchases made on credit over assets.

6.6 Table 6: Trade-credit demand

Read:

- The dependent variable is accounts payable over assets.
- Predicted supply of trade credit is positive and statistically significant across specifications.
- Positive sales growth is associated with higher accounts payable, consistent with demand for financing when firms expand.
- Net profits over assets enters negatively, indicating that firms with weaker internal funds use more trade credit.
- Available bank credit enters negatively, and longer lender relationships are associated with lower accounts payable.
- Current assets excluding cash enter positively, consistent with working-capital needs.
- Price variables for early-payment discounts provide weaker evidence than the quantity and credit-availability variables.

Economic message: Firms use trade credit when suppliers make it available and when their own financing needs are high. Trade credit is especially important when internal funds and bank credit are limited.

Table 6
Trade credit demand: Accounts payable over assets

Independent variable	I	II ^b	III ^c	IV	V
Supply of trade credit					
Predicted supply of trade credit/assets ^a	.027** (.013)	.029** (.013)	.085* (.019)	.027** (.013)	.028** (.013)
Demand for capital					
Δ sales (86–87)/assets	.012* (.003)	–.001 (.008)	.012* (.003)	.012* (.003)	.012* (.003)
if positive, zero otherwise					
Δ sales (86–87)/assets	–.004 (.005)	.007 (.006)	–.002 (.005)	–.005 (.005)	–.004 (.005)
if negative, zero otherwise					
1986 sales missing (0,1)	.002 (.014)	–.001 (.017)	.002 (.014)	.002 (.014)	.005 (.014)
Log(book value of assets)	.005 (.003)	.004 (.003)	–.002 (.003)	.006** (.003)	.004 (.003)
Log(1+firm age)	.029*** (.016)	.036*** (.019)	.023 (.017)	.029*** (.016)	.029*** (.016)
Log(1+firm age) ²	–.006*** (.004)	–.006 (.004)	–.004 (.004)	–.006 (.004)	–.006 (.004)
Current assets excluding cash/assets	.166* (.013)	.156* (.018)	.168* (.015)	.169* (.013)	.161* (.013)
Firm applied for loan during previous year (0,1)	.002 (.010)	.005 (.011)	.003 (.010)	.003 (.010)	.001 (.010)
Credit availability					
Net profits/assets	–.019* (.006)	–.021* (.007)	–.039* (.007)	–.019* (.006)	–.019* (.006)
Available line of credit/assets				–.063* (.022)	
Firm denied credit request during previous year (0,1)	.012 (.017)	.019 (.019)	.010 (.017)	.012 (.017)	.012 (.017)
Log(1+longest relationship with lender) (in years)	–.011** (.005)	–.018* (.006)	–.009*** (.005)	–.011* (.005)	–.013* (.005)
Firm is located in an MSA	.017** (.007)	.012* (.008)	.013*** (.007)	.017*** (.007)	.019* (.007)
Price of trade credit					
Percent of credit purchases offered with early payment discounts					.006 (.011)
Early payment discounts missing					–.035* (.010)
Number of observations	1968	1588	1968	1968	1968
R ²	14.9	20.0	17.6	15.3	15.6

The dependent variable is the accounts payable:asset ratio reported by the firm. The coefficients are estimated using ordinary least squares. Standard errors are in parentheses. Each regression includes five industry dummy variables and a constant whose coefficients are not reported. Firms in the financial industry (SIC 60000–6999) and service industry (SIC 7000–8999) are excluded unless otherwise noted. Low asset values also caused some of the independent variables to be skewed. When the distribution of a variable was highly skewed, we recoded the upper percent of values to the 99th percentile of the distribution. These variables include purchase on account to assets, operating profits to assets, employees to assets, and number of trade creditors to assets. ^aPredicted trade credit supply is the estimated value of purchase on account over assets from Table 5, column 1. We adjusted our estimate of purchases on account by subtracting off 0.200 (the coefficient on employment over assets) times the value of employment over assets for each observation. This removed the mismeasurement in purchases on account caused by wages being included in cost of goods sold, but not purchases on account.

^bThis regression was estimated with instrumental variables. In addition to the other variables in the regression, we used employees per thousand dollars of assets, whether the firm was incorporated, and the percent of inventory which is finished goods as instruments. These are the variables that we used in Table 5 to estimate trade credit supply.

^cThis regression includes 36 industry dummies, one for each two-digit SIC code in our dataset.

*, **, ***Coefficient significantly different from zero at the 1%, 5%, and 10% levels, respectively.

Table 6: Trade-credit demand: accounts payable over assets.

7 Short summary

- The paper studies why firms use trade credit instead of only borrowing from financial institutions.
- Trade credit has two sides: accounts receivable are credit offered to customers, and accounts payable are credit received from suppliers.
- The paper tests financing-advantage, price-discrimination, and transaction-cost theories.

- Small-firm evidence comes from the National Survey of Small Business Finances; large-firm comparison evidence comes from Compustat.
- Larger and financially stronger firms extend more trade credit to customers.
- Firms with more available bank credit extend more credit, consistent with suppliers using their own financing capacity to fund customers.
- Credit purchases over assets are used to proxy for trade-credit supply from suppliers.
- Suppliers offer more credit to firms that look more creditworthy or more valuable as customers.
- Suppliers offer less credit when the goods are harder to recover or liquidate, as proxied by the share of inventory that is finished goods.
- Accounts payable are higher when predicted supply is higher, when firms are growing, when internal funds are weaker, and when access to bank credit is poorer.
- Trade credit is therefore both a financing substitute and a product-market relationship tool.

8 Conclusion

8.1 Core conclusion

Petersen and Rajan show that trade credit is economically important because suppliers have advantages that financial institutions sometimes lack. Firms with better access to capital markets extend trade credit, and firms with weaker access to institutional finance use it. The evidence supports a financing-advantage interpretation, while also leaving room for price discrimination, customer-relationship, and transaction-cost motives.

8.2 Economic intuition

A supplier knows more about its customer than a distant lender may know. It observes orders, payment behavior, inventory use, and product-market prospects. It can also threaten to stop future deliveries and may be better able to recover or resell goods. These advantages allow suppliers to extend credit when banks are reluctant, especially to smaller firms.

8.3 Incremental contribution revisited

The paper's key contribution is to turn theories of trade credit into testable empirical predictions. It uses detailed small-firm data to separately study the firm as a supplier of credit and as a user of credit. This separation is what allows the paper to distinguish supply-side and demand-side explanations.

8.4 Implications

Trade credit should be viewed as part of the financial system. It links firms together through short-term credit chains and can substitute for institutional finance. This matters for small-business finance, working-capital management, and the propagation of shocks across firms.

8.5 Limitations

The evidence is observational and based on proxies. Accounts payable and receivable are imperfect measures of desired borrowing and lending. Trade terms vary across industries, and repayment speed is not always observed. The small-firm survey is rich but limited in time, while the large-firm comparison uses a different data source.

8.6 Takeaway

The enduring takeaway is that suppliers are not merely sellers of goods; they are also lenders. Trade credit exists because suppliers sometimes have the information, control, and relationship advantages needed to finance customers when banks or capital markets cannot do so as effectively.